# Practice workbook · Calculating the range

The range as a measure of dispersion.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-18

# 1. Question 3 · part a · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part a only.

**3** Davina investigated the relationship between the time spent on psychology homework and end of topic test scores.

Davina asked her students to write down the average time (in minutes) they spent on their homework each week and she also recorded their end of topic test scores (out of 30 marks).

Davina's results are shown in **Table 3**.

| Participant | Average time spent on homework (in minutes) | End of topic test scores (out of 30 marks) |
|---|---|---|
| **A** | 50 | 17 |
| **B** | 25 | 10 |
| **C** | 30 | 15 |
| **D** | 20 | 10 |
| **E** | 55 | 25 |
| **F** | 15 | 9 |
| **G** | 80 | 30 |
| **H** | 50 | 19 |
| **I** | 60 | 27 |
| **J** | 20 | 14 |

**Table 3**

(a) Calculate the range for the **end of topic test scores**.

**(1)**

Range .....

(b) Calculate how many students spent **more than** 30 minutes on their homework as a fraction of all students.

You must give your answer in the lowest form.

**(2)**

Fraction .....

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P 7 8 7 3 4 A 0 8 5 2

## 1. Question 3 · part a · 1 mark · Working and answer

Working and answer

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# 2. Question 2 · part c · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part c only.

**2** Michelle conducted a study to find out if 15 year olds had better long-term memory recall than 55 year olds. She gave two groups of participants one hour to memorise a set of 40 pictures. After two weeks, she asked the participants to recall as many of the pictures as they could.

Her results are shown in **Table 1**.

| 15 year olds | Number of pictures recalled correctly | 55 year olds | Number of pictures recalled correctly |
|---|---|---|---|
| Participant A | 35 | Participant A | 16 |
| Participant B | 30 | Participant B | 28 |
| Participant C | 29 | Participant C | 27 |
| Participant D | 18 | Participant D | 34 |
| Participant E | 38 | Participant E | 35 |
| Participant F | 31 | Participant F | 26 |
| Participant G | 24 | Participant G | 21 |

**Table 1**

(a) Calculate the mean score for the number of pictures recalled by 55 year olds.

You must give your answer to two decimal places.

**(2)**

Mean score for 55 year olds .....

(b) Calculate the median score for the number of pictures recalled by 15 year olds.

**(2)**

Median score for 15 year olds .....

(c) Calculate the range for the number of pictures recalled by all participants.

**(1)**

Range for all participants .....

**(Total for Question 2 = 5 marks)**

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## 2. Question 2 · part c · 1 mark · Working and answer

Working and answer

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## 3. Question 3 · part c · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part c only.

**3** Ryan used a questionnaire to assess whether students had a fixed or growth mindset in four secondary schools. He asked the students to complete the questionnaire online.

His results are shown in **Table 2**.

| School | Number of students with a fixed mindset | Number of students with a growth mindset |
|---|---|---|
| A | 458 | 931 |
| B | 211 | 439 |
| C | 641 | 838 |
| D | 189 | 271 |

**Table 2**

(a) Calculate the mean number of students with a fixed mindset.

**(1)**

Mean .....

(b) Calculate the total number of students who completed the questionnaire.

**(1)**

Total .....

(c) Calculate the range for the number of students with a growth mindset.

**(1)**

Range .....

**(Total for Question 3 = 3 marks)**

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## 3. Question 3 · part c · 1 mark · Working and answer

Working and answer

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# 4. Question 4 · part d · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part d only.

**4** Jamie investigated whether personality influences obedience to an authority figure. He used a questionnaire with open-ended and closed-ended questions about personality traits and obedient behaviour.

(a) Describe **one** reason why Jamie may have used open-ended questions in his investigation.

**(2)**

(b) State **one** ethical issue that Jamie should have considered in his investigation.

**(1)**

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# 4. Question 4 · part d · 1 mark

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Further practice · Attempt part d only.

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Jamie found that eight of his participants scored highly for authoritarian personality.

The number of obedient acts they had reported for the previous week are shown in **Table 4**.

| Participant | Number of reported obedient acts (in a week) |
|---|---|
| **A** | 15 |
| **B** | 14 |
| **C** | 18 |
| **D** | 16 |
| **E** | 18 |
| **F** | 14 |
| **G** | 13 |
| **H** | 19 |

**Table 4**

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(c) Calculate the median score for the number of obedient acts reported by participants in **Table 4**.

**(2)**

Median .....

(d) Calculate the range for the number of obedient acts reported by the participants in **Table 4**.

**(1)**

Range .....

**(Total for Question 4 = 6 marks)**

## 4. Question 4 · part d · 1 mark · Working and answer

Working and answer

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## 5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

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**1** Gary conducted an experiment to test whether there was a difference in the ability of boys and girls to conserve number. He sampled 50 boys and 50 girls aged six years old.

Gary showed them the image in **Figure 1**.

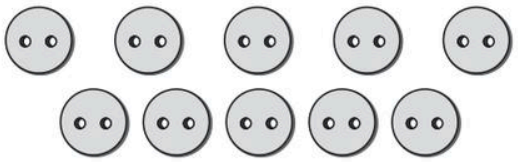


**Figure 1**

Gary asked each child to say whether the top row of buttons contained more, less, or the same number of buttons as the bottom row.

(a) Identify the independent variable in Gary's experiment.

**(1)**

- ☒ **A** age
- ☒ **B** gender
- ☒ **C** ability
- ☒ **D** conservation

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(b) Give **two** ways that Gary could ensure his experiment is ethical.

**(2)**

1 .....

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2 .....

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## 5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

The results of Gary's experiment are shown in **Table 1**.

| | Number of children who said the top row contained less buttons than the bottom row | Number of children who said the top row contained the same number of buttons as the bottom row | Number of children who said the top row contained more buttons than the bottom row |
|---|---|---|---|
| **Boys** | 5 | 13 | 32 |
| **Girls** | 1 | 25 | 24 |

**Table 1**

(c) Calculate the percentage of girls who said that the top row contained the **same** number of buttons as the bottom row, out of all the children in the experiment.

**(2)**

Percentage .....

(d) Calculate the range for the number of children who said the top row contained **more** buttons than the bottom row.

**(1)**

Range .....

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P 7 2 3 8 3 A 0 4 4 8

## 5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

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(e) Calculate the ratio for the number of children who said that the top row contained **more** buttons than the bottom row, to all children taking part in the experiment.

**(2)**

Ratio .....

(f) Explain **one** conclusion that Gary could make about the ability to conserve number between boys and girls from the results in **Table 1**.

**(2)**

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**(Total for Question 1 = 10 marks)**

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Working and answer

## 6. Question 3 · part a · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a only.

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**3** Savannah is investigating whether there is a relationship between listening to music and happiness. She asked participants to record the number of hours they spent listening to music in a week. They also scored their level of happiness on a scale of 1 (being very unhappy) to 10 (being very happy).

Savannah sampled a total of 180 participants aged between 18 years old and 65 years old. A third of her sample were aged between 18 years old and 25 years old.

(a) Calculate the range for the age of all participants.

**(1)**

Range .....

(b) Calculate the ratio of participants aged between 18 years old and 25 years old to the remaining participants in the sample used by Savannah.

**(2)**

Ratio .....

(c) Savannah found that those listening to less than one hour of music in a week scored themselves at 2 or below on the happiness scale. Happiness scores increased for those who listened to more music, with those who listened to over 12 hours of music in a week usually scoring 9 or above.

Explain **one** conclusion that Savannah could make from her findings about the relationship between music and happiness.

**(2)**

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**(Total for Question 3 = 5 marks)**

## 6. Question 3 · part a · 1 mark · Working and answer

Working and answer

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